26900187 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCECT301, MG5DSCECC301 - ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. What is the main purpose of real-time problem-solving in AI? [U] / [CO1]

   a). To solve problems only after collecting all data

   b). To make decisions within a limited time while responding to changing situations

   c). To avoid making decisions

   d). To store information permanently

2. Identify which connective is represented by A → B? [K] / [CO2]

   a). Negation b). Conjunction

   c). Implication d). Biconditional

3. In SVM, what are "support vectors"? [K] / [CO2, CO3, CO4]

a). All training data points used to train the model.

b). The data points that lie exactly on the margin boundaries.

c). The data points that are misclassified by the model.

d). The features with the highest variance.

4. Why are searching algorithms important in AI problem-solving? [U] / [CO1]

a). They help an agent explore possible solutions to reach a goal

b). They eliminate the need for input data

c). They only store information

d). They increase the size of the search space

5. Recognize the Learning type which learns from mistakes or successes through feedback is associated with: [K] / [CO2]

a). Supervised Learning b). Unsupervised Learning

c). Reinforcement Learning d). Classification

6. What is a 'Local Optimum' in the context of Local Search? [K] / [CO1]

a). A state whose neighbors have lower objective function values than the current state.

b). A state whose neighbors have higher objective function values than the current state.

c). A state whose neighbors have equal objective function values.

d). The absolute best state in the entire search space.

7. What is Adversarial Search? [U] / [CO2]

a). A search technique used when an AI competes against another agent

b). A technique for sorting data

c). A database searching technique

d). A file searching technique

8. In fraud detection, fraudulent transactions are very rare compared with normal transactions. Why can outlier detection be useful? [A] / [CO3]

a). Fraudulent transactions may differ significantly from normal transactions

b). Fraudulent transactions are always missing values

c). Fraudulent transactions are always nominal

d). Outliers cannot be used in machine learning

9. Why should an AI agent identify its goal before solving a problem? [U] / [CO1]

a). To know the desired final state

b). To reduce the size of the computer

c). To avoid using algorithms

d). To increase the number of actions

10. Show what is the first distribution from which rejection sampling generates samples? [K] / [CO2]

a). Posterior distribution b). Prior distribution

c). Utility distribution d). Conditional distribution only

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. What is crossover? Explain its role in Genetic Algorithms. [U] / [CO1]

12. Explain the Naïve Bayes model. [U] / [CO2]

13. What is the basic difference between Bagging and Boosting? [U] / [CO2, CO3, CO4]

14. Describe the definition and scope of Artificial Intelligence with suitable examples. [U] / [CO1]

15. Compare L1 and L2 regularization methods. [U] / [CO3]

16. Explain the concept of problem formulation in Artificial Intelligence. Describe the important components involved in formulating an AI problem with a suitable example. [U] / [CO1]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Classify independence and conditional independence in probability-based intelligent agents. [A] / [CO2]

18. Construct and illustrate a confusion matrix. Classify and show TP, TN, FP, and FN, and compute the four main evaluation parameters. [A] / [CO3]

19. Explain the A* search algorithm in detail. Describe the role of g(n), h(n), and f(n), and explain how A* selects the next node for expansion. [U] / [CO1]

20. Explain Alpha-Beta pruning in detail. Discuss its need, working principle, Alpha and Beta values, and pruning condition. [An] / [CO2]

[10x2 = 20]